

BANDIT : Over The Wire

```
(kali㉿kali)-[~]
└─$ ssh bandit0@bandit.labs.overthewire.org -p 2220
```



LEVEL : **bandit0 → bandit1**

The password for the next level is stored in a file called **readme** located in the home directory. Use this password to log into bandit1 using SSH. Whenever you find a password for a level, use SSH (on port 2220) to log into that level and continue the game.

```
bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If
```

Passwd : **ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If**

LEVEL : **bandit1 → bandit2**

Connect to the server of bandit1 before solving this using :

```
ssh bandit1@bandit.labs.overthewire.org -p 2220
```

The password for the next level is stored in a file called - located in the home directory.

```
bandit1@bandit:~$ ls
-
bandit1@bandit:~$ ls -la
total 24
-rw-r----- 1 bandit2 bandit1 33 Apr 3 15:17 -
drwxr-xr-x 2 root root 4096 Apr 3 15:17 .
drwxr-xr-x 150 root root 4096 Apr 3 15:20 ..
-rw-r--r-- 1 root root 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 root root 3851 Apr 3 15:10 .bashrc
-rw-r--r-- 1 root root 807 Mar 31 2024 .profile
bandit1@bandit:~$ cat ./-
263JGJPfgU6LtdEvgfWU1XP5yac29mFx
bandit1@bandit:~$
```

Passwd : **263JGJPfgU6LtdEvgfWU1XP5yac29mFx**

LEVEL : **bandit2 → bandit3**

The password for the next level is stored in a file called `--spaces in this filename--` located in the home directory.

```
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ ls -al
total 24
drwxr-xr-x 2 root root 4096 Apr 3 15:17 .
drwxr-xr-x 150 root root 4096 Apr 3 15:20 ..
-rw-r--r-- 1 root root 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 root root 3851 Apr 3 15:10 .bashrc
-rw-r--r-- 1 root root 807 Mar 31 2024 .profile
-rw-r----- 1 bandit3 bandit2 33 Apr 3 15:17 --spaces in this filename--
bandit2@bandit:~$ cat -- "--spaces in this filename--"
MNk8KNH3Usiio41PRUEoDFPqfxLPISmx
bandit2@bandit:~$
```

Passwd : **MNk8KNH3Usiio41PRUEoDFPqfxLPISmx**

LEVEL : **bandit3 → bandit4**

The password for the next level is stored in a hidden file in inhere directory.

```
bandit3@bandit:~$ ls -al
total 24
drwxr-xr-x  3 root root 4096 Apr  3 15:17 .
drwxr-xr-x 150 root root 4096 Apr  3 15:20 ..
-rw-r--r--  1 root root  220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root root 3851 Apr  3 15:10 .bashrc
drwxr-xr-x  2 root root 4096 Apr  3 15:17 inhere
-rw-r--r--  1 root root  807 Mar 31 2024 .profile
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -al
total 12
drwxr-xr-x 2 root  root  4096 Apr  3 15:17 .
drwxr-xr-x 3 root  root  4096 Apr  3 15:17 ..
-rw-r----- 1 bandit4 bandit3  33 Apr  3 15:17 ... Hiding-From-You
bandit3@bandit:~/inhere$ cat "... Hiding-From-You"
2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
bandit3@bandit:~/inhere$
```

Passwd : **2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ**

LEVEL : **bandit4 → bandit5**

The password for the next level is stored in the only human-readable file in the **inhere** directory. Tip: if your terminal is messed up, try the "reset" command.

```
bandit4@bandit:~$ ls inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls -al
total 48
drwxr-xr-x 2 root    root    4096 Apr  3 15:17 .
drwxr-xr-x 3 root    root    4096 Apr  3 15:17 ..
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file00
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file01
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file02
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file03
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file04
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file05
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file06
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file07
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file08
-rw-r----- 1 bandit5 bandit4   33 Apr  3 15:17 -file09
bandit4@bandit:~/inhere$ file ./-file07 | grep "text"
./-file07: ASCII text
bandit4@bandit:~/inhere$ cat ./-file07
4oQYVPkxZ00E005pTW81FB8j8lxXGUQw
```

Passwd : **4oQYVPkxZ00E005pTW81FB8j8lxXGUQw**

LEVEL : bandit5 → bandit6

The password for the next level is stored in a file somewhere under the **inhere** directory and has all of the following properties:

- human-readable
- 1033 bytes in size
- not executable

```
bandit5@bandit:~/inhere$ ls -R
.: Home st Downloads
maybehere00 maybehere03 maybehere06 maybehere09 maybehere12 maybehere15 maybehere1
maybehere01 maybehere04 maybehere07 maybehere10 maybehere13 maybehere16 maybehere1
maybehere02 maybehere05 maybehere08 maybehere11 maybehere14 maybehere17

./maybehere00:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3
File System
./maybehere01:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere02:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere03:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere04:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere05:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere06:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere07:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere08:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere09:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere10:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere11:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere12:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere13:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere14:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere15:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3

./maybehere16:
-file1 -file2 -file3 spaces file1 spaces file2 spaces file3
```

```
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable  
./maybehere07/.file2  
bandit5@bandit:~/inhere$ cat ./maybehere07/.file2  
HWasnPhtq9AVKe0dmk45nxy20cvUa6EG  
  
bandit5@bandit:~/inhere$
```

Passwd : **HWasnPhtq9AVKe0dmk45nxy20cvUa6EG**

LEVEL : **bandit6 → bandit7**

The password for the next level is stored **somewhere on the server** and has all of the following properties:

- owned by user bandit7
- owned by group bandit6
- 33 bytes in size

```
bandit6@bandit:~$ find / -type f -user bandit7 -group bandit6 -size 33c 2>/dev/null  
/var/lib/dpkg/info/bandit7.password  
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password  
morbNTDkSW6jIlUc0ym0dMaLn0LFVAaj
```

Passwd : **morbNTDkSW6jIlUc0ym0dMaLn0LFVAaj**

LEVEL : **bandit7 → bandit8**

The password for the next level is stored in the file **data.txt** next to the word **millionth**.

```
bandit7@bandit:~$ ls
data.txt
bandit7@bandit:~$ cat data.tx | grep "millionth"
cat: data.tx: No such file or directory
bandit7@bandit:~$ cat data.txt | grep "millionth"
millionth      dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc
bandit7@bandit:~$
```

Passwd : **dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc**

LEVEL : **bandit8 → bandit9**

The password for the next level is stored in the file **data.txt** and is the only line of text that occurs only once.

```
bandit8@bandit:~$ ls -l
total 36
-rw-r----- 1 bandit9 bandit8 33033 Apr  3 15:18 data.txt
bandit8@bandit:~$ sort data.txt | uniq -u
4CKMh1JI91bUIZZPXDqGanal4xvAg0JM
bandit8@bandit:~$
```

Passwd : **4CKMh1JI91bUIZZPXDqGanal4xvAg0JM**

LEVEL : **bandit9 → bandit10**

The password for the next level is stored in the file **data.txt** in one of the few human-readable strings, preceded by several '=' characters.

```
bandit9@bandit:~$ ls -al
total 40
drwxr-xr-x  2 root    root    4096 Apr  3 15:17 .
drwxr-xr-x 150 root    root    4096 Apr  3 15:20 ..
-rw-r--r--  1 root    root     220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root   3851 Apr  3 15:10 .bashrc
-rw-r-----  1 bandit10 bandit9 19382 Apr  3 15:17 data.txt
-rw-r--r--  1 root    root     807 Mar 31 2024 .profile
bandit9@bandit:~$ strings data.txt | grep "=" | tail -n 1
]\{=
bandit9@bandit:~$ strings data.txt | grep "="
===== the
I\=Ow
V?L=
%3=VZ
===== password
={M\
===== is
=Dvq
=n/N
===== FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey
zX]% =
]\{=
bandit9@bandit:~$
```

Passwd : **FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey**

LEVEL : bandit10 → bandit11

The password for the next level is stored in the file **data.txt**, which contains base64 encoded data.

```
bandit10@bandit:~$ ls -l
total 4
-rw-r----- 1 bandit11 bandit10 69 Apr  3 15:17 data.txt
bandit10@bandit:~$ ls -al
total 24
drwxr-xr-x  2 root    root    4096 Apr  3 15:17 .
drwxr-xr-x 150 root    root    4096 Apr  3 15:20 ..
-rw-r--r--  1 root    root     220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root   3851 Apr  3 15:10 .bashrc
-rw-r-----  1 bandit11 bandit10  69 Apr  3 15:17 data.txt
-rw-r--r--  1 root    root     807 Mar 31 2024 .profile
bandit10@bandit:~$ cat data.txt
VGhlIHBhc3N3b3JkIGlzIGR0UjE3M2ZaS2IwUlJzREZTR3NnMlJXbnBOVmozczVJYcg=
bandit10@bandit:~$ echo "VGhlIHBhc3N3b3JkIGlzIGR0UjE3M2ZaS2IwUlJzREZTR3NnMlJXbnBOVmozczVJYcg=" | base64 -d
The password is dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr
bandit10@bandit:~$
```

Passwd : **dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr**

LEVEL : bandit11 → bandit12

The password for the next level is stored in the file **data.txt**, where all lowercase (a-z) and uppercase (A-Z) letters have been rotated by 13 positions.

```
bandit11@bandit:~$ cat data.txt | tr 'A-Za-z' 'N-ZA-Mn_za-m'  
Tmm zmmmm__m mm 7m16WNmHIm5YmImWmmFIa__mnUTmm9Q4  
bandit11@bandit:~$ cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'  
The password is 7x16WNeHIi5YkIhWsFIqoognUTyj9Q4  
bandit11@bandit:~$
```

Passwd : 7x16WNeHIi5YkIhWsFIqoognUTyj9Q4

LEVEL : bandit12 → bandit13

The password for the next level is stored in the file **data.txt**, which is a hexdump of a file that has been repeatedly compressed. For this level it may be useful to create a directory under /tmp in which you can work. Use mkdir with a hard to guess directory name. Or better, use the command "mktemp -d". Then copy the datafile using cp, and rename it using mv (read the manpages!)

```

bandit12@bandit:~$ mkdir /tmp/der12
bandit12@bandit:~$ cd /tmp/der12
bandit12@bandit:/tmp/der12$ cp ~/data.txt /tmp/der12/
bandit12@bandit:/tmp/der12$ xxd -r data.txt > data.bin
bandit12@bandit:/tmp/der12$ file data.bin
data.bin: gzip compressed data, was "data2.bin", last modified: Fri Apr 3 15:17:36 2026, max compression, from Unix, original size modulo 2^32 576
bandit12@bandit:/tmp/der12$ mv data.bin data.gz
gzip -d data.gz # produces data (or original name)
bandit12@bandit:/tmp/der12$ gzip -d data.gz
gzip: data.gz: No such file or directory
bandit12@bandit:/tmp/der12$ ls
data data.txt
bandit12@bandit:/tmp/der12$ file data
data: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/der12$ mv data data.bz2
bandit12@bandit:/tmp/der12$ bzip2 -d data.bz2
bandit12@bandit:/tmp/der12$ ls
data data.txt
bandit12@bandit:/tmp/der12$ file data
data: gzip compressed data, was "data4.bin", last modified: Fri Apr 3 15:17:36 2026, max compression, from Unix, original size modulo 2^32 20480
bandit12@bandit:/tmp/der12$ mv data data.gz
gzip -d data.gz
ls
file data4.bin
data data.txt
data4.bin: cannot open `data4.bin' (No such file or directory)
bandit12@bandit:/tmp/der12$ ls
data data.txt
bandit12@bandit:/tmp/der12$ file data
data: POSIX tar archive (GNU)
bandit12@bandit:/tmp/der12$ mv data data.tar
bandit12@bandit:/tmp/der12$ tar -xf data.tar
bandit12@bandit:/tmp/der12$ ls
data5.bin data.tar data.txt
bandit12@bandit:/tmp/der12$ file data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/der12$ mv data5.bin data.tar
bandit12@bandit:/tmp/der12$ tar -xf data.tar
bandit12@bandit:/tmp/der12$ ls
data6.bin data.tar data.txt
bandit12@bandit:/tmp/der12$ mv data6.bin data.tar
bandit12@bandit:/tmp/der12$ tar -xf data.tar
bandit12@bandit:/tmp/der12$ ls
data8.bin data.tar data.txt
bandit12@bandit:/tmp/der12$ file data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Fri Apr 3 15:17:36 2026, max compression, from Unix, original size modulo 2^32 49
bandit12@bandit:/tmp/der12$ file data8.bin bin data.gz
data8.bin bin: cannot open `data8.bin bin' (No such file or directory)
data.gz: cannot open `data.gz' (No such file or directory)
bandit12@bandit:/tmp/der12$ mv data8.bin data.gz
bandit12@bandit:/tmp/der12$ gzip -d data.gz

```

```

bandit12@bandit:/tmp/der12$ file data8.bin bin data.gz
data8.bin bin: cannot open `data8.bin bin' (No such file or directory)
data.gz: cannot open `data.gz' (No such file or directory)
bandit12@bandit:/tmp/der12$ mv data8.bin data.gz
bandit12@bandit:/tmp/der12$ gzip -d data.gz
bandit12@bandit:/tmp/der12$ ls
data data.tar data.txt
bandit12@bandit:/tmp/der12$ file data
data: ASCII text
bandit12@bandit:/tmp/der12$ cat data
The password is F05dwFsc0cbaIiH0h8J2eUks2vdTDwAn

```

Passwd : **F05dwFsc0cbaIiH0h8J2eUks2vdTDwAn**

LEVEL : **bandit13 → bandit14**

The password for the next level is stored in **/etc/bandit_pass/bandit14** and can only be read by user **bandit14**. For this level, you don't get the next password, but you get a private SSH key that can be used to log into the next level. Look

at the commands that logged you into previous bandit levels, and find out how to use the key for this level.

```
bandit13@bandit:~$ ls
HINT  sshkey.private
bandit13@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.

(dermicool@ beach)-[~]
$ scp -P 2220 bandit13@bandit.labs.overthewire.org:/home/bandit13/sshkey.private .

      _ _ _ _ _
     / / / / /
    / / / / /
   / / / / /
  / / / / /
 / / / / /
/ / / / /

This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit13@bandit.labs.overthewire.org's password:
sshkey.private                                100% 1679      2.7KB/s   00:00

(dermicool@ beach)-[~]
$ ls
BurpSuiteCommunity  Documents  hashlib    Pictures    Python-2.7.18.tgz  Videos
crime_mount         Downloads  int.py     pow.solver.py solve.py          'VirtualBox VMs'
crime_scene.img     exploit.py itertools  sshkey.private sshkey.private    volatility
Desktop             get-pip.py Music       Python-2.7.18  Templates

(dermicool@ beach)-[~]
$ chmod 600 sshkey.private

(dermicool@ beach)-[~]
$ ssh -i sshkey.private bandit14@bandit.labs.overthewire.org -p 2220

      _ _ _ _ _
     / / / / /
    / / / / /
   / / / / /
  / / / / /
 / / / / /
/ / / / /
```

```
Enjoy your stay!

bandit14@bandit:~$ ls
bandit14@bandit:~$ ls -al
total 24
drwxr-xr-x    3 root root 4096 Apr  3 15:17 .
drwxr-xr-x 150 root root 4096 Apr  3 15:20 ..
-rw-r--r--    1 root root  220 Mar 31  2024 .bash_logout
-rw-r--r--    1 root root 3851 Apr  3 15:10 .bashrc
-rw-r--r--    1 root root  807 Mar 31  2024 .profile
drwxr-xr-x    2 root root 4096 Apr  3 15:17 .ssh
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14
MU4VWeTyJk8R0of1qqmcBPALh7lDCPvS
```

Passwd : MU4VWeTyJk8R0of1qqmcBPALh7lDCPvS

LEVEL : bandit14 → bandit15

The password for the next level can be retrieved by submitting the password of the current level to **port 30000 on localhost**.

```
Session Actions Edit View Help
bandit14@bandit:~$ telnet localhost 30000
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
MU4VWeTyJk8R0of1qqmcBPALh7lDCPvS
Correct!
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo

Connection closed by foreign host.
bandit14@bandit:~$
```

Passwd : **8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo**

LEVEL : bandit15 → bandit16

The password for the next level can be retrieved by submitting the password of the current level to **port 30001 on localhost** using SSL/TLS encryption.

```
bandit15@bandit:~$ ls
bandit15@bandit:~$ echo '8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo' | openssl s_client -quiet -connect localhost:30001
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
Correct!
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx
```

Passwd : **kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx**

LEVEL : bandit16 → bandit17

The credentials for the next level can be retrieved by submitting the password of the current level to **a port on localhost in the range 31000 to 32000**. First find out which of these ports have a server listening on them. Then find out

which of those speak SSL/TLS and which don't. There is only 1 server that will give the next credentials, the others will simply send back to you whatever you send to it.

```
bandit16@bandit:~$ nmap -vv -p31000-32000 -sV localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-20 09:09 UTC
NSE: Loaded 46 scripts for scanning.
Initiating Ping Scan at 09:09
Scanning localhost (127.0.0.1) [2 ports]
Completed Ping Scan at 09:09, 0.00s elapsed (1 total hosts)
Initiating Connect Scan at 09:09
Scanning localhost (127.0.0.1) [1001 ports]
Discovered open port 31960/tcp on 127.0.0.1
Discovered open port 31046/tcp on 127.0.0.1
Discovered open port 31691/tcp on 127.0.0.1
Discovered open port 31518/tcp on 127.0.0.1
Discovered open port 31790/tcp on 127.0.0.1
Completed Connect Scan at 09:09, 0.03s elapsed (1001 total ports)
Initiating Service scan at 09:09
Scanning 5 services on localhost (127.0.0.1)
Service scan Timing: About 20.00% done; ETC: 09:12 (0:02:40 remaining)
Completed Service scan at 09:11, 132.65s elapsed (5 services on 1 host)
NSE: Script scanning 127.0.0.1.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 09:11
Completed NSE at 09:11, 0.01s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 09:11
Completed NSE at 09:11, 0.01s elapsed
Nmap scan report for localhost (127.0.0.1)
Host is up, received conn-refused (0.00016s latency).
Scanned at 2026-04-20 09:09:20 UTC for 133s
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE      REASON  VERSION
31046/tcp  open  echo         syn-ack
31518/tcp  open  ssl/echo     syn-ack
31691/tcp  open  echo         syn-ack
31790/tcp  open  ssl/unknown  syn-ack
31960/tcp  open  echo         syn-ack
```

```

bandit16@bandit:~$ echo 'kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx' | openssl s_client -quiet -connect localhost:31790
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
Correct!
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAvM0kuiFmMg6HL2YPI0jon6iWfbp7c3jx34YkYWqUH57SUdyJ
imZzeyGC0gtZPGujUSxiJSWI/oTqexh+cAMTSMl0Jf7+BrJ0bArnxd9Y7YT2bRPQ
Ja6Lzb558YW3FZl870Ri0+rW4LDCdNd2lUvLE/GL2GwyuKN0K5iCd5TbtJzEkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WWAW18BR6yGrMq7Q/kALHYW30ekePQAZL0VUYbW
JGTi65CxbCnzc/w4+mqQyvmzpWtMAzJTzAzQxNbK2MBGySxDLrjg0LWN6sK7wNX
x0YVztz/zbIkPjfkU1jHS+9EbVNj+D1XF0JuaQIDAQABAOIBABagpxpM1aoLWfvD
KHcj10nqcoBc4oE11aFYQwik7xfW+24pRNUDE6SFth0ar69jp5RlLwD1NhP
x3iBl
J9n0M80J0VToum43U0S8YxF8WwhXriYGnc1sskbwpX0UDc9uX4+UESzH22P29ovd
d8WErY0gPxun8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQn
Pw9nC
-----END RSA PRIVATE KEY-----

```

PRIVATE SSH KEY :

```
-----BEGIN RSA PRIVATE KEY-----
```

```

MIIEogIBAAKCAQEAvM0kuiFmMg6HL2YPI0jon6iWfbp7c3jx34YkYWqUH57
SUdyJ
imZzeyGC0gtZPGujUSxiJSWI/oTqexh+cAMTSMl0Jf7+BrJ0bArnxd9Y7YT
2bRPQ
Ja6Lzb558YW3FZl870Ri0+rW4LDCdNd2lUvLE/GL2GwyuKN0K5iCd5TbtJz
EkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WWAW18BR6yGrMq7Q/kALHYW30ekePQAZL0
VUYbW
JGTi65CxbCnzc/w4+mqQyvmzpWtMAzJTzAzQxNbK2MBGySxDLrjg0LWN6s
K7wNX
x0YVztz/zbIkPjfkU1jHS+9EbVNj+D1XF0JuaQIDAQABAOIBABagpxpM1ao
LWfvD
KHcj10nqcoBc4oE11aFYQwik7xfW+24pRNUDE6SFth0ar69jp5RlLwD1NhP
x3iBl
J9n0M80J0VToum43U0S8YxF8WwhXriYGnc1sskbwpX0UDc9uX4+UESzH22P
29ovd
d8WErY0gPxun8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQn
Pw9nC

```



```
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52y0Q9q0kwFTEQpj tF4uNtJom+asvl  
pmS8A  
vLY9r60wYSvmZhNqBUrj7lyCtXMIu1kkd4w7F77k+DjHoAXyxcUp1DGL51s  
Omama  
+TOWWgECgYEA8JtPxP0GRJ+IQkX262jM3dEI kza8ky5moIwUqYdsx0NxHgR  
Rh0RT  
8c8hAuRBb2G82so8vUHK/fur850Efc9TncnCY2crpoqsg hifKLxrLgtT+qD  
pfZnx  
SatLdt8GfQ85yA7hnWWJ2MxF3NaeSDm75Lsm+tBbAiy c9P2jGRNtMSkCgYE  
AypHd  
HCctNi/FwjulhttFx/rHYKhLidZDFYeiE/v45bN4yFm8x7R/b0iE7KaszX+  
Exdvt  
SghaTdcG0Knyw1bpJVyusavPzpaJMjdJ6tcFhVAbAjm7enCIvGCSx+X3l5S  
iWg0A  
R57hJglezIiVjv3aGwHwvlZvtszK6zV6oXFAu0ECgYAbjo46T4hyP5tJi93  
V5HDi  
Ttiek7xRVxUl+iU7rWkGAXFpMLFteQEsRr7PJ/lemmEY5eTDAFMLy9FL2m9  
oQWCg  
R8VdwSk8r9FGLS+9aKcV5PI/WEKlwgXinB30hYimtiG2Cg5JCqIZFHxD6Mj  
EG0iu  
L8ktHMPvodBwNsSBULpG0QKBgBAplTfC1H0nWiMG0U3KPwYwt006CdTkmJO  
mL8Ni  
blh9elyZ9FsGxsgtRBXRsqXuz7wtsQAgLHxbdLq/ZJQ7Yfz0KU4ZxEnabvX  
nvWkU  
Y0djHdS0oKvDQNWu6ucyLRAWFuISeXw9a/9p7ftpxm0TSgyvmfLF2MIAEwy  
zRqaM  
77pBAoGAMmj mIJdjp+Ez8duyn3ieo36yr ttf5NSsJLABxFpdlc1gvtGCWW+  
9Cq0b  
dxviW8+TFVEBl104f7HVm6EpTscdDxU+bCXWkfjuRb7Dy9G0tt9JP sX8MBT  
akzh3  
vBgysi/sN3RqRBcGU40f0oZyfAMT8s1m/uYv5206IgeuZ/ujbjY=  
-----END RSA PRIVATE KEY-----
```

LEVEL : **bandit17** → **bandit18**

There are 2 files in the **homedirectory**: **passwords.old** and **passwords.new**.
The password for the next level is in **passwords.new** and is the only line that

has been changed between **passwords.old** and **passwords.new**.

```
(dermicool@beach)-[~]  
$ nano bandit17.key  
  
(dermicool@beach)-[~]  
$ ssh -i bandit17.key bandit17@bandit.labs.overthewire.org -p 2220
```

Paste the private key in bandit17.key.

After pasting, change permissions using `chmod 600 bandit17.key` & then log into level 17 using :

bandit17@bandit.labs.overthewire.org -p 2220

```
bandit17@bandit:~$ diff passwords.old passwords.new
42c42
< KxOU4IzbXM8j8HeAWPAXTd1eC77mp1qV
—
> x2gLTTjFwMOhQ8oWNbMN362QKxfRqGLO
```

Passwd : x2gLTTjFwM0hQ8oWNbMN362QKxfRqGL0

LEVEL : bandit18 → bandit19

The password for the next level is stored in a file **readme** in the homedirectory. Unfortunately, someone has modified **.bashrc** to log you out when you log in with SSH.

```
(dermicool@beach)-[~]
$ ssh bandit18@bandit.labs.overthewire.org -p 2220 "cat readme"

      _ _ _ _ _
     /   /   /
    /___/___/___\
   /___/___/___\
  /___/___/___\
 /___/___/___\
/___/___/___\

      This is an OverTheWire game server.
      More information on http://www.overthewire.org/wargames

backend: gibbon-1
bandit18@bandit.labs.overthewire.org's password:
cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8
```

Passwd : cGwPmaKXVwDUNgPAVJbWYuGHVn9z13j8

LEVEL : bandit19 → bandit20

To gain access to the next level, you should use the setuid binary in the homedirectory. Execute it without arguments to find out how to use it. The password for this level can be found in the usual place (/etc/bandit_pass), after you have used the setuid binary.

```
bandit19@bandit:~$ ls
bandit20-do  st_ Downloads
bandit19@bandit:~$ ls -la
total 36
drwxr-xr-x  2 root    root      4096 Apr  3 15:17 .
drwxr-xr-x 150 root    root      4096 Apr  3 15:20 ..
-rwsr-x---  1 bandit20 bandit19 14888 Apr  3 15:17 bandit20-do
-rw-r--r--  1 root    root       220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root      3851 Apr  3 15:10 .bashrc
-rw-r--r--  1 root    root       807 Mar 31 2024 .profile
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbY0
```

Passwd : 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbY0

LEVEL : bandit20 → bandit21

```
bandit20@bandit:~$ echo '0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbY0' | nc -nlvp 1234 6
[1] 31
bandit20@bandit:~$ Listening on 0.0.0.0 1234
netstat -tulpn
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
netstat: no support for 'AF_INET (tcp)' on this system.
tcp6      0      0 :::31960                :::*                     LISTEN      -
tcp6      0      0 :::22                   :::*                     LISTEN      -
tcp6      0      0 :::2223                 :::*                     LISTEN      -
tcp6      0      0 :::2221                 :::*                     LISTEN      -
tcp6      0      0 :::2220                 :::*                     LISTEN      -
tcp6      0      0 :::2232                 :::*                     LISTEN      -
tcp6      0      0 :::2231                 :::*                     LISTEN      -
tcp6      0      0 :::2230                 :::*                     LISTEN      -
tcp6      0      0 :::2228                 :::*                     LISTEN      -
tcp6      0      0 :::2227                 :::*                     LISTEN      -
tcp6      0      0 :::2226                 :::*                     LISTEN      -
tcp6      0      0 :::2225                 :::*                     LISTEN      -
tcp6      0      0 :::2224                 :::*                     LISTEN      -
tcp6      0      0 :::31046                 :::*                     LISTEN      -
tcp6      0      0 :::31691                 :::*                     LISTEN      -
bandit20@bandit:~$ la
.bash_logout .bashrc .profile suconnect
bandit20@bandit:~$ ./suconnect 1234
Connection received on 127.0.0.1 59642
Read: 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbY0
Password matches, sending next password
EeoULMCra2q0dSkYj561DX7s1CpBu0Bt
[1]+  Done                  echo '0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbY0' | nc -nlvp 1234
```

Passwd : EeoULMCra2q0dSkYj561DX7s1CpBu0Bt

LEVEL : bandit21 → bandit22

A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in **/etc/cron.d/** for the configuration and see what command is being executed.

```
bandit21@bandit:~$ ls
bandit21@bandit:~$ ls /etc/cron.d/
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup  otw-tmp-dir
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job  sysstat
bandit21@bandit:~$ cat /etc/cron.d/cronjob_bandit22
@reboot bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
* * * * * bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
bandit21@bandit:~$ cat /usr/bin/cronjob_bandit22.sh
#!/bin/bash
chmod 644 /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
cat /etc/bandit_pass/bandit22 > /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
bandit21@bandit:~$ cat /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q
```

Passwd : **tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q**

LEVEL : bandit22 → bandit23

A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in **/etc/cron.d/** for the configuration and see what command is being executed.

NOTE: Looking at shell scripts written by other people is a very useful skill. The script for this level is intentionally made easy to read. If you are having problems understanding what it does, try executing it to see the debug information it prints.

```
bandit22@bandit:~$ la
.bash_logout .bashrc .profile
bandit22@bandit:~$ ls /etc/cron.d
behemoth4_cleanup cronjob_bandit22 cronjob_bandit24 leviathan5_cleanup otw-tmp-dir
clean_tmp cronjob_bandit23 e2scrub_all manpage3_resetpw_job sysstat
bandit22@bandit:~$ cat /etc/cron.d/cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
bandit22@bandit:~$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash

myname=$(whoami)
mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)

echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"

cat /etc/bandit_pass/$myname > /tmp/$mytarget
bandit22@bandit:~$ echo 'I am user bandit23' | md5sum
8ca319486bfbbc3663ea0f8e81326349 -
bandit22@bandit:~$ cat /tmp/8ca319486bfbbc3663ea0f8e81326349
0Zf11ioIjMVN551jX3CmStKLYqjk54Ga
bandit22@bandit:~$
```

Passwd : **0Zf11ioIjMVN551jX3CmStKLYqjk54Ga**

LEVEL : **bandit23 → bandit24**

A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in **/etc/cron.d/** for the configuration and see what command is being executed.

```
bandit22@bandit:~$ ls
bandit22@bandit:~$ ls /etc/cron.d
behemoth4_cleanup cronjob_bandit22 cronjob_bandit24 leviathan5_cleanup otw-tmp-dir
clean_tmp cronjob_bandit23 e2scrub_all manpage3_resetpw_job sysstat
bandit22@bandit:~$ cat /etc/cron.d/cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null
bandit22@bandit:~$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash

shopt -s nullglob
myname=$(whoami)

cd /var/spool/"$myname"/foo || exit
echo "Executing and deleting all scripts in /var/spool/$myname/foo:"
for i in *; do
    if [ "$i" != "." ] && [ "$i" != ".." ]; then
        echo "Handling $i"
        name=$(stat -format "%N" "$i")
        if [ "$name" = "bandit23" ] && [ -f "$i" ]; then
            timeout -s 900 "$i"
            fi
            rm -rf "$i"
        fi
    done
bandit22@bandit:~$ nano /var/spool/bandit24/foo/dernicol.sh
Unable to create directory /home/bandit24/.local/share/nano: No such file or directory
It is required for saving/loading search history or cursor positions.
bandit22@bandit:~$ donebandit23@bandit:~$ nano /var/spool/bandit24/foo/dernicol.sh
donebandit23@bandit:~$ command not found
bandit22@bandit:~$ nano /var/spool/bandit24/foo/dernicol.sh
Unable to create directory /home/bandit24/.local/share/nano: No such file or directory
It is required for saving/loading search history or cursor positions.
bandit22@bandit:~$ chmod +x /var/spool/bandit24/foo/dernicol.sh
bandit22@bandit:~$ cat /var/spool/bandit24/foo/dernicol.sh
bandit22@bandit:~$ echo 'cat /etc/bandit_pass/bandit24' > /tmp/dernicol_bandit24_pass.txt
bandit22@bandit:~$ cat /tmp/dernicol_bandit24_pass.txt
bandit22@bandit:~$ cat /tmp/dernicol_bandit24_pass.txt
gb8KRRCsshZXI0tUuR6yp0Fjizbf3G8
bandit22@bandit:~$
```

Passwd : **gb8KRRCsshZXI0tUuR6yp0Fjizbf3G8**

LEVEL : **bandit24 → bandit25**

A daemon is listening on port 30002 and will give you the password for bandit25 if given the password for bandit24 and a secret numeric 4-digit

pincode. There is no way to retrieve the pincode except by going through all of the 10000 combinations, called brute-forcing.

```
bandit24@bandit:~$ mkdir -p /tmp/banshit
cd /tmp/banshit
bandit24@bandit:/tmp/banshit$ nano brute.sh
Unable to create directory /home/bandit24/.local/share/nano/: No such file or directory
It is required for saving/loading search history or cursor positions.

bandit24@bandit:/tmp/banshit$ chmod +x brute.sh
./brute.sh
I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a single line, separated by a space.
Correct!
The password of user bandit25 is iCi86ttT4KSNe1armKiwbQNmB3YJP3q4
```

nano brute.sh :

```
#!/bin/bash

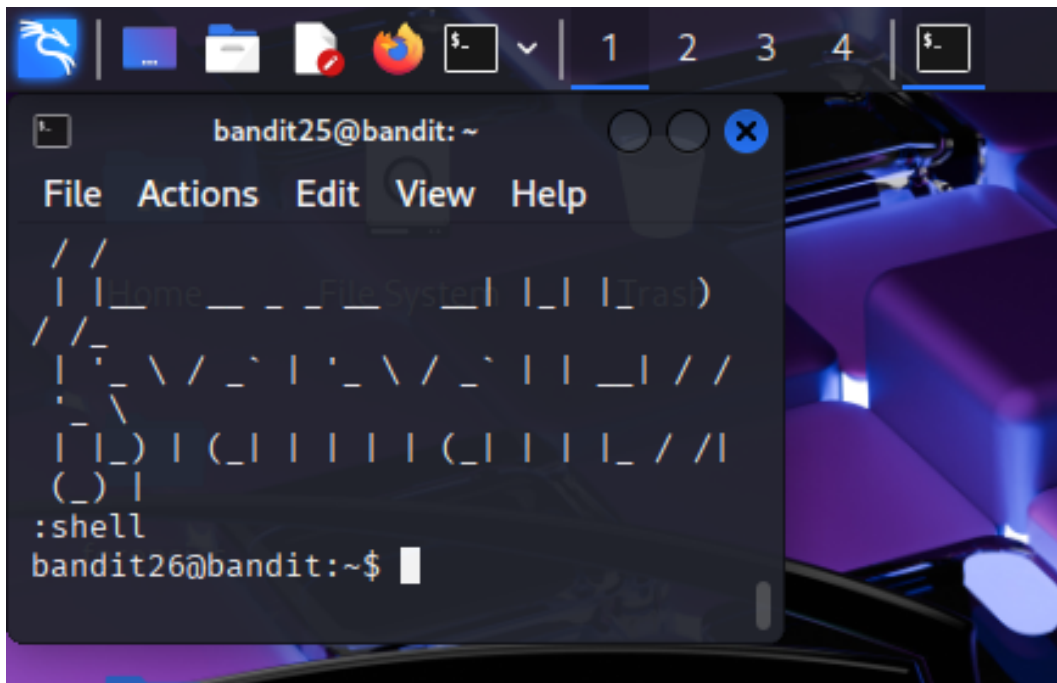
password="gb8KRRcsshZXI0tUuR6ypOFjiZbf3G8"

for i in $(seq -w 0 9999); do
    echo "$password $i"
done | nc localhost 30002 | grep -v "Wrong" | grep -v "Enter"
```

Passwd : **iCi86ttT4KSNe1armKiwbQNmB3YJP3q4**

LEVEL : **bandit25 → bandit26**

Logging in to bandit26 from bandit25 should be fairly easy... The shell for user bandit26 is not **/bin/bash**, but something else. Find out what it is, how it works and how to break out of it.



Passwd : `s0773xxkk0MXfdq0fPRVr9L3jJBU0gCZ`

LEVEL : `bandit26 → bandit27`

Good job getting a shell! Now hurry and grab the password for bandit27.

```
bandit26@bandit:~$ ls
bandit27-do  text.txt
bandit26@bandit:~$ find / -user bandit27 2>/dev/null
/etc/bandit_pass/bandit27
/home/bandit26/bandit27-do
bandit26@bandit:~$ ./bandit27-do cat /etc/bandit_pass/bandit27
/home/bandit26/bandit27-do
upsNCc7vzaRDx6oZC6GiR6ERwe1MowGB
```

Passwd : **upsNCc7vzaRDx6oZC6GiR6ERwe1MowGB**

LEVEL : **bandit27 → bandit28**

There is a git repository at `ssh://bandit27-git@bandit.labs.overthewire.org/home/bandit27-git/repo` via the port `2220`. The password for the user `bandit27-git` is the same as for the user `bandit27`.

```
(dermicool@beach)-[~]
$ mkdir /tmp/myrepo
cd /tmp/myrepo

(dermicool@beach)-[/tmp/myrepo]
$ git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo
Cloning into 'repo' ...
File System
Trash
This is an OverTheWire game server.
More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit27-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (3/3), done.

(dermicool@beach)-[/tmp/myrepo]
$ cd repo

(dermicool@beach)-[/tmp/myrepo/repo]
$ cat README
The password to the next level is: Yz9IpL0sBcCeuG7m9uQFt8ZNpS4HZRcN
```

Passwd : **Yz9IpL0sBcCeuG7m9uQFt8ZNpS4HZRcN**

LEVEL : **bandit28 → bandit29**

There is a git repository at `ssh://bandit28-git@bandit.labs.overthewire.org/home/bandit28-git/repo` via the port `2220`. The

Passwd : **4pT1t5DENaYuqnqvadYs1oE4QLCdjmJ7**

LEVEL : **bandit29 → bandit30**

There is a git repository at `ssh://bandit29-git@bandit.labs.overthewire.org/home/bandit29-git/repo` via the port `2220`. The password for the user `bandit29-git` is the same as for the user `bandit29`.

From your local machine (not the OverTheWire machine!), clone the repository and find the password for the next level. This needs git installed locally on your machine.

```
(dermicool@beach)~$ mkdir /tmp/myrepo29
$ cd /tmp/myrepo29
$ git clone ssh://bandit29-git@bandit.labs.overthewire.org:2220/home/bandit29-git/repo
$ cd repo
Cloning into 'repo' ...
```

```
(dermicool@beach)-[/tmp/myrepo29/repo]
$ git branch -a
* master
  remotes/origin/HEAD → origin/master
  remotes/origin/dev
  remotes/origin/master
  remotes/origin/sploits-dev

(dermicool@beach)-[/tmp/myrepo29/repo]
$ git checkout dev
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'

(dermicool@beach)-[/tmp/myrepo29/repo]
$ cat README.md
# Bandit Notes
Some notes for bandit30 of bandit.

## credentials

- username: bandit30
- password: qp30ex3VLz5MDG1n91YowTv4Q8l7CDZL
```

Passwd : **qp30ex3VLz5MDG1n91YowTv4Q8l7CDZL**

LEVEL : bandit30 → bandit31

There is a git repository at `ssh://bandit30-`

`git@bandit.labs.overthewire.org/home/bandit30-git/repo` via the port `2220`. The password for the user `bandit30-git` is the same as for the user `bandit30`.

From your local machine (not the OverTheWire machine!), clone the repository and find the password for the next level. This needs git installed locally on your machine

```
(dermicool@beach)-[~]
$ mkdir /tmp/myrepo30
cd /tmp/myrepo30
git clone ssh://bandit30-git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo
cd repo
mkdir: cannot create directory '/tmp/myrepo30': File exists
fatal: destination path 'repo' already exists and is not an empty directory.

(dermicool@beach)-[/tmp/myrepo30/repo]
$ cat README.md
git log -p
git branch -a
just an empty file... muahaha
commit e761c5d8e8d01ea9567e35cafe0d58e0e540397a (HEAD → master, origin/master, origin/HEAD)
Author: Ben Dover <noone@overthewire.org>
Date: Fri Apr 3 15:17:41 2026 +0000

    initial commit of README.md

diff --git a/README.md b/README.md
new file mode 100644
index 0000000..029ba42
--- /dev/null
+++ b/README.md
@@ -0,0 +1 @@
+just an empty file... muahaha
* master
remotes/origin/HEAD → origin/master
remotes/origin/master

(dermicool@beach)-[/tmp/myrepo30/repo]
$ git tag
secret

(dermicool@beach)-[/tmp/myrepo30/repo]
$ git show secret
fb5S2xb7bRyFmAvQYQGEqsbhVyJqhnDy
```

Passwd : `fb5S2xb7bRyFmAvQYQGEqsbhVyJqhnDy`

LEVEL : bandit31 → bandit32

There is a git repository at `ssh://bandit31-`

`git@bandit.labs.overthewire.org/home/bandit31-git/repo` via the port `2220`. The password for the user `bandit31-git` is the same as for the user `bandit31`.

From your local machine (not the OverTheWire machine!), clone the repository and find the password for the next level. This needs git installed locally on your machine.

#FLOW OF COMMANDS#

```
mkdir /tmp/myrepo31
cd /tmp/myrepo31
git clone ssh://bandit31-git@bandit.labs.overthewire.org:22
20/home/bandit31-git/repo
cd repo

git config --global user.email "GMAIL"
git config --global user.name "USERNAME"

echo "May I come in?" > key.txt

git add -f key.txt

git commit -m "push key"
git push origin master
```

Passwd : 309RfhqyALVBEZpVb6LYStshZoqoSx5K